

Chapter 1

Automap

Automaper is a python script that converts a simple Excel description of a photonic chip into a complete measurement plan ready for automated characterization.

Instead of hardcoding device coordinates or manually programming measurement routines, the user only needs to describe the chip structure in an Excel file then `automaper.py` interprets this information and automatically generates the data structures required by the positioning software.

The Excel file is organized in independent sheets, where each sheet describes a different aspect of the chip as follows:

Chapter 2

Instalación TwinCAT XAE eXtended Automation Engineering

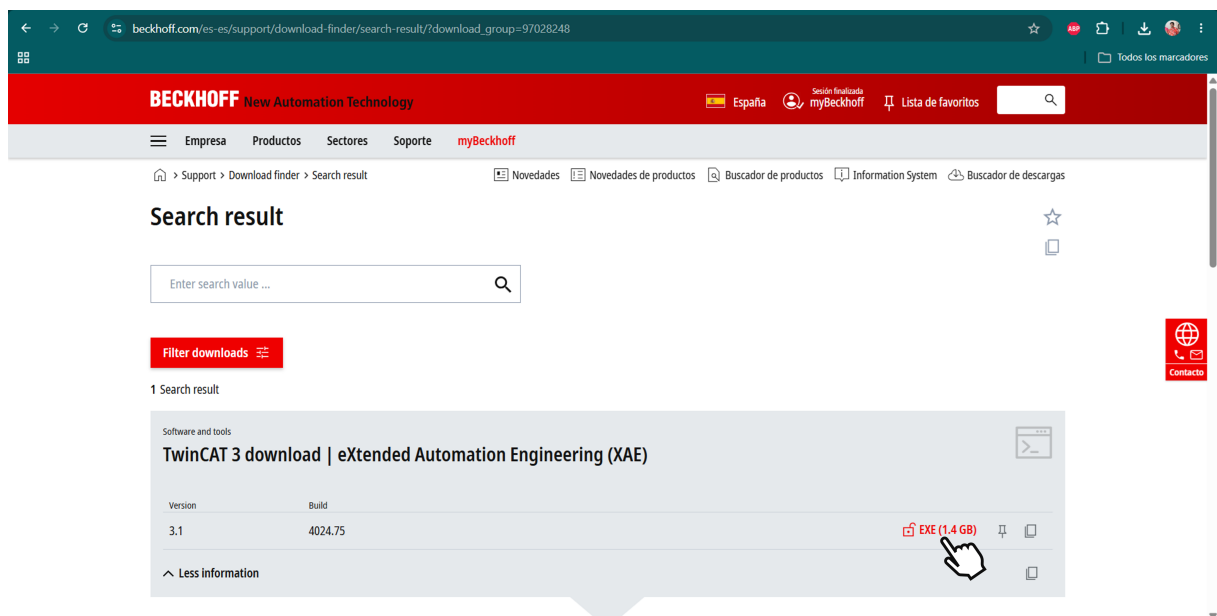


Figure 2.1: Pendiente.

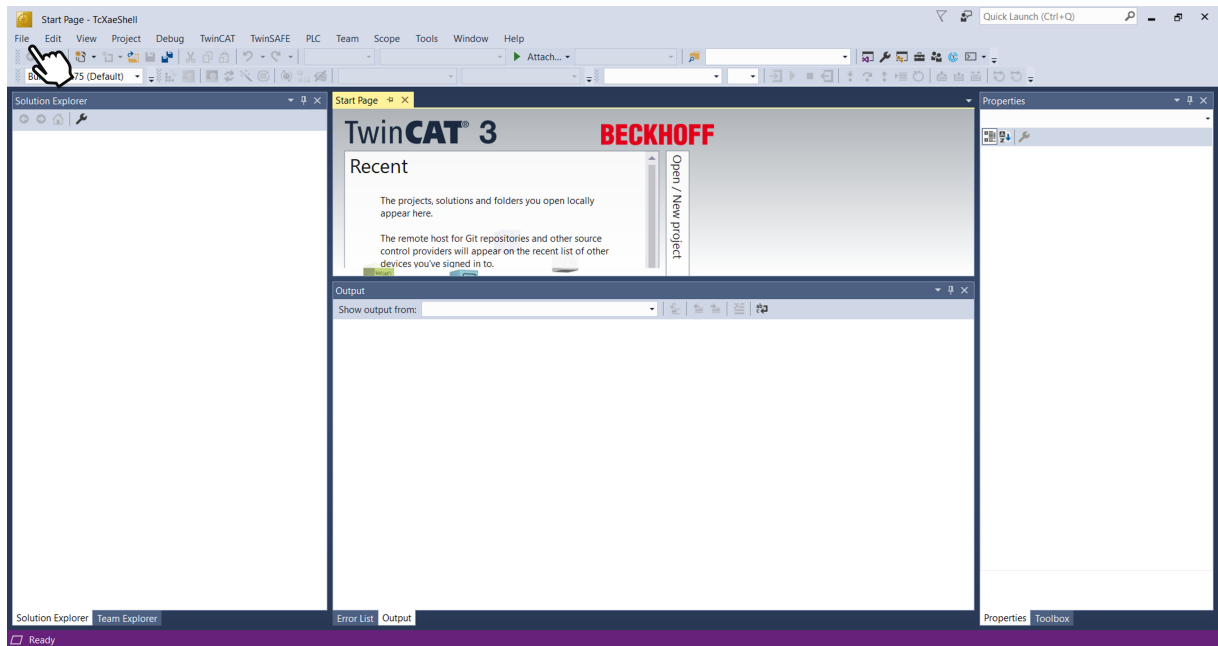


Figure 2.2: Pendiente.

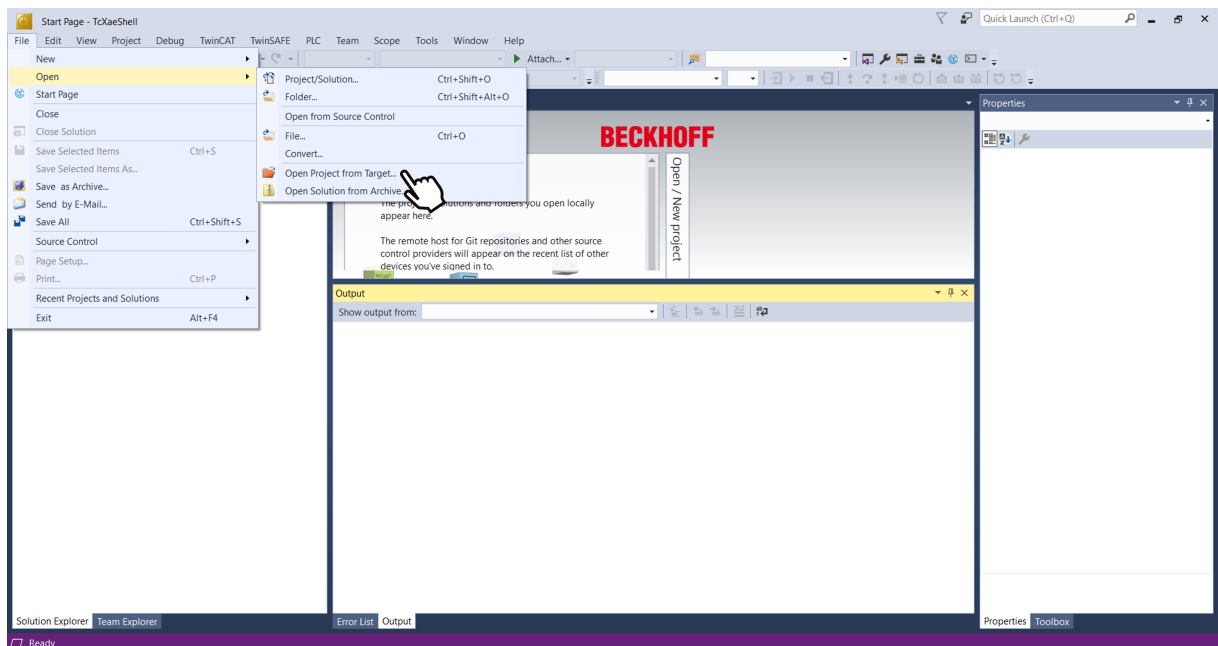


Figure 2.3: Pendiente.

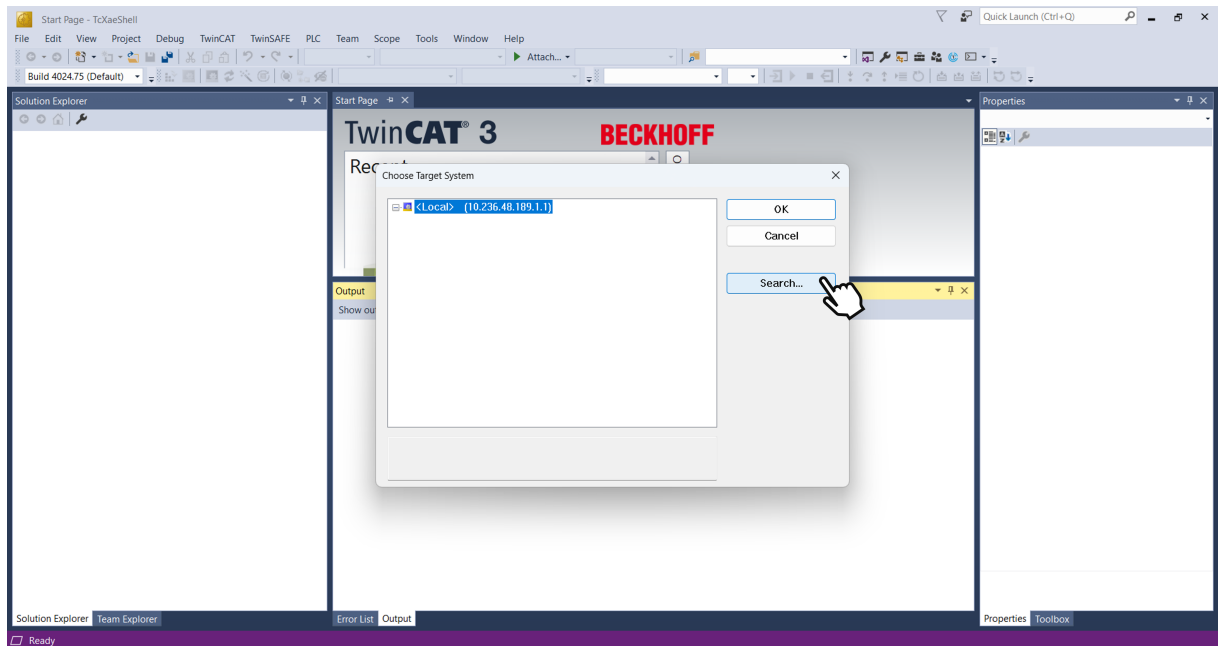


Figure 2.4: Pendiente.

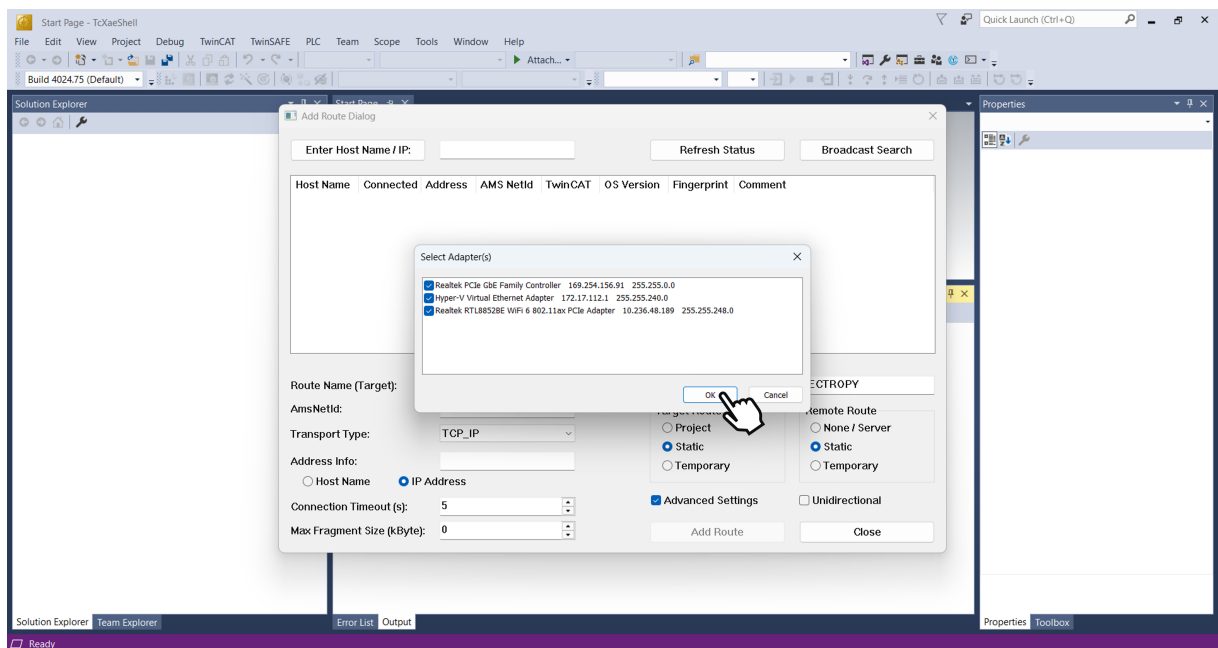


Figure 2.5: Pendiente.

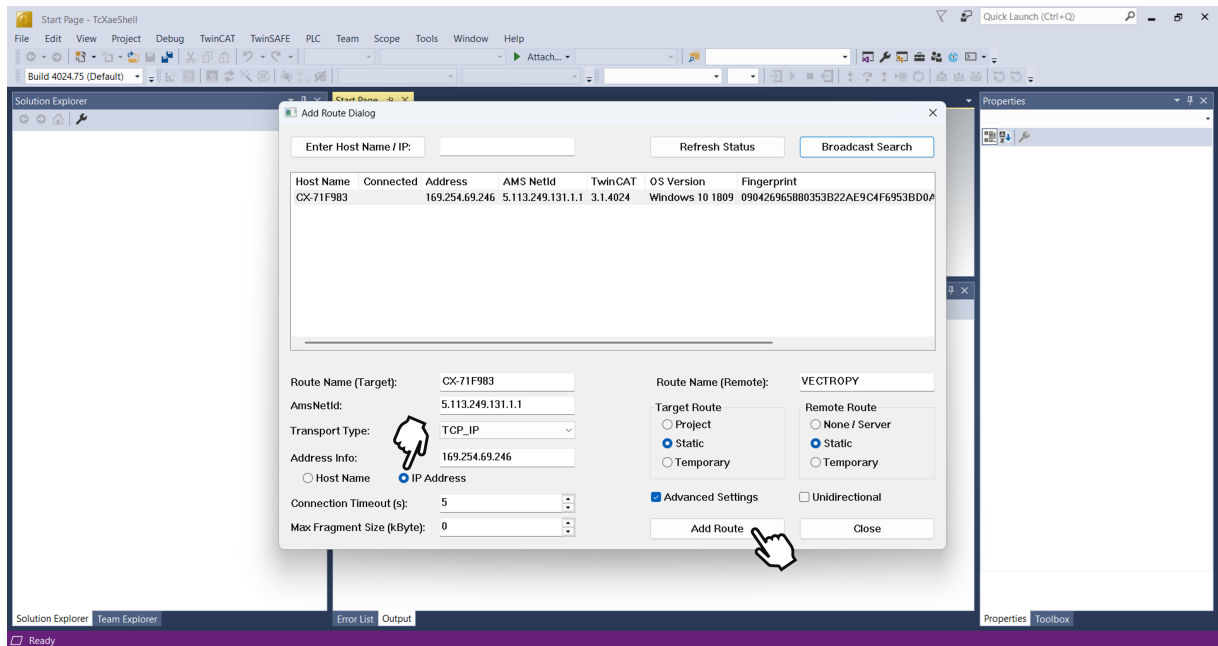


Figure 2.6: Pendiente.

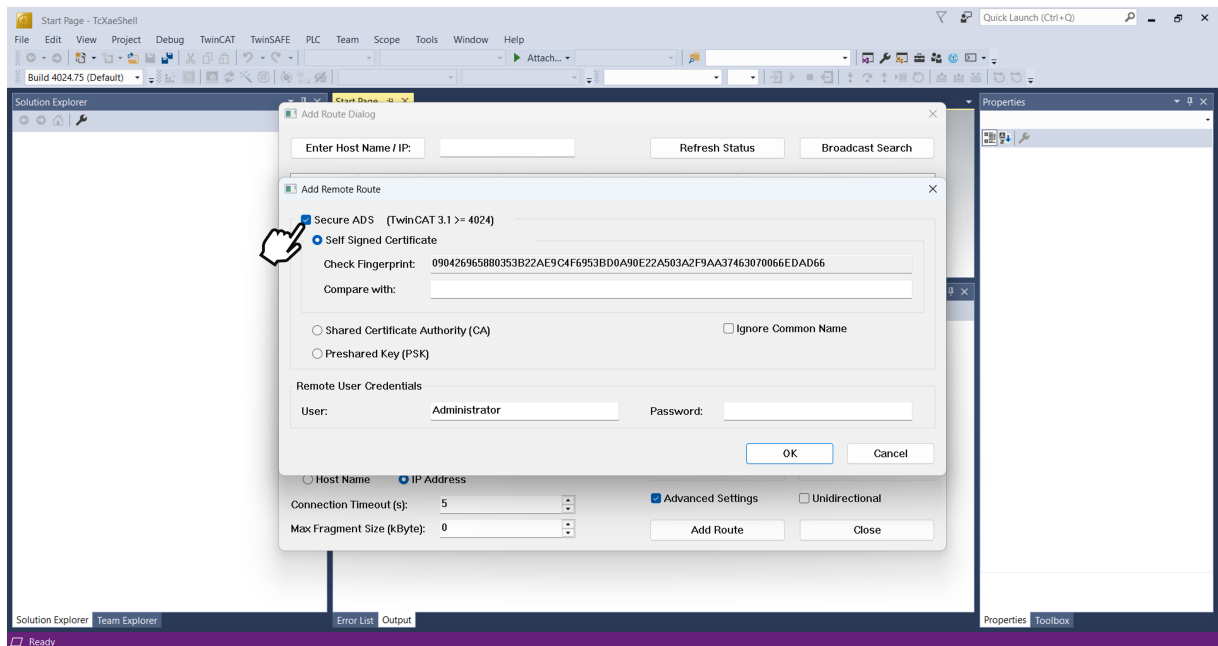


Figure 2.7: Pendiente.

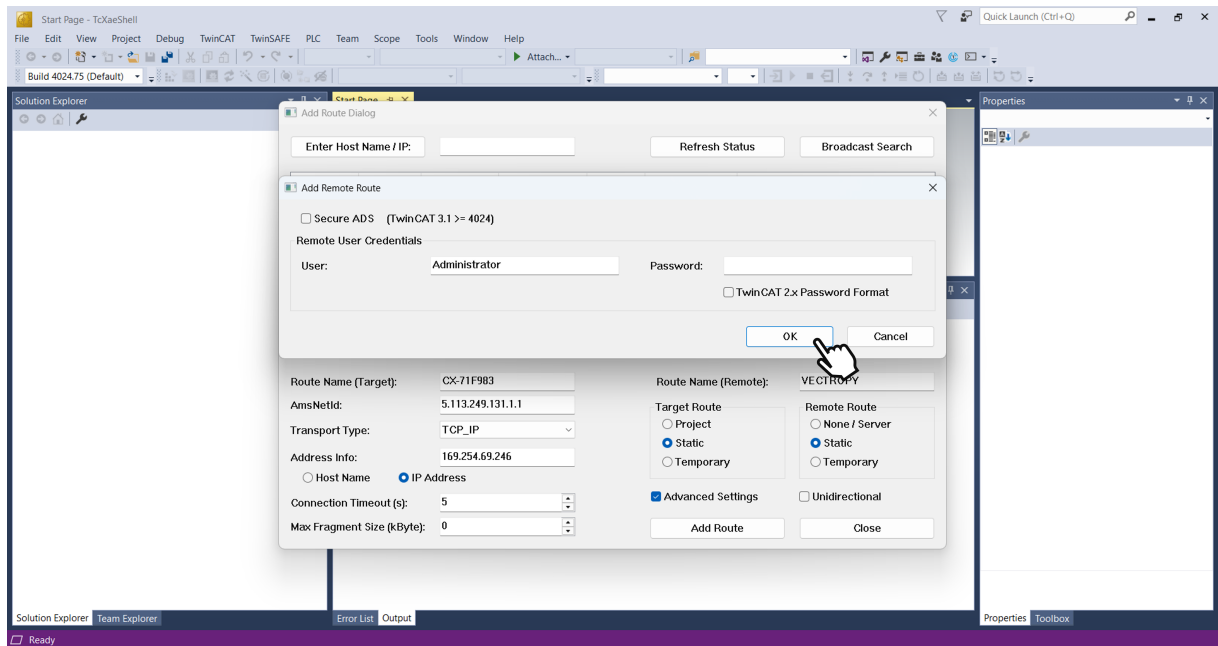


Figure 2.8: Pendiente.

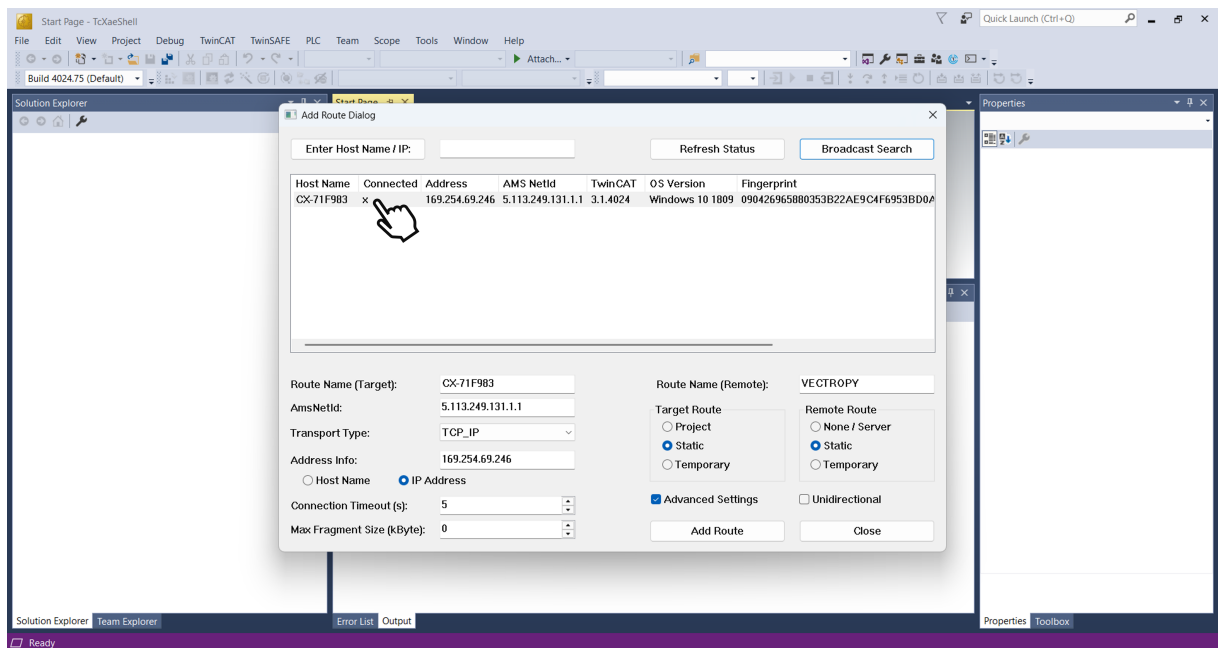


Figure 2.9: Pendiente.

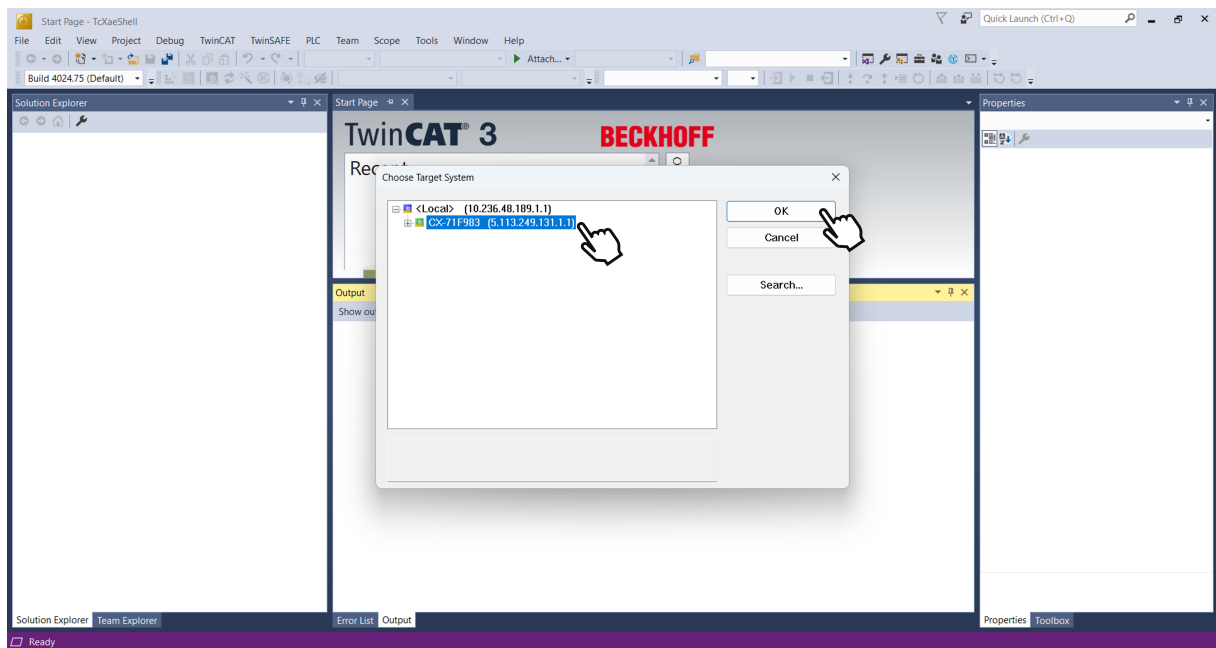


Figure 2.10: Pendiente.

Chapter 3

Documentacion de la API Suruga Seiki

Es necesario cargar el ensamblado (assembly), una unidad de código previamente compilada que permite establecer la comunicación con la plataforma Suruga Seiki. Este ensamblado está compuesto por un conjunto de clases que definen las propiedades (atributos) y los métodos (acciones o funcionalidades) disponibles en la plataforma.

En programación orientada a objetos, los métodos representan las acciones que una clase puede ejecutar, mientras que las propiedades describen las características asociadas a sus objetos. Un objeto es una instancia concreta de una clase; utilizando una analogía, la clase puede entenderse como un plano de construcción y el objeto como el resultado material obtenido a partir de ese plano.

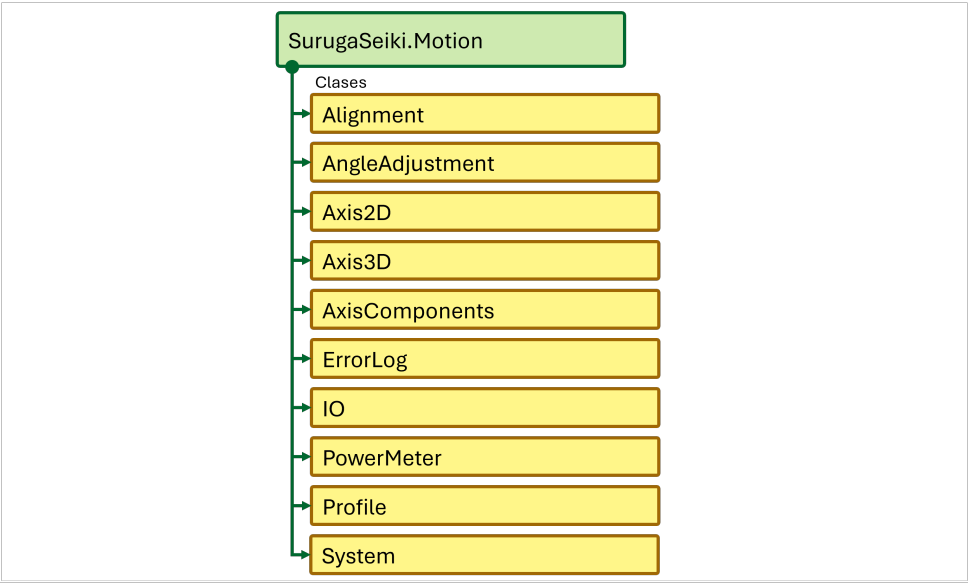


Figure 3.1: Pendiente.

3.1 Clase Alignment

```
=====
SurugaSeiki.Motion.Alignment
=====

Constructores:
```

```

Alignment()
Alignment(UInt16 id)

Metodos:
    ErrorCode StartSingle()
    ErrorCode StartFlat()
    ErrorCode StartFocus()
    ErrorCode StartTwoChRotation()
    ErrorCode StartMMFRotation()
    ErrorCode StartLoopBack()
    ErrorCode StartCommon()
    ErrorCode Stop()
    ErrorCode GetErrorCode()
    Status GetStatus()
    UInt32 GetErrorAxisID()
    AligningStatusCode GetAligningStatus()
    UInt32 GetProfilePacketSumIndex(ProfileDataType profileDataType)
    ProfileData RequestProfileData(ProfileDataType profileDataType, UInt32 index)
    ProfileData RequestProfileData(ProfileDataType profileDataType, UInt32 index, Boolean isRotationCenterShift)
    Double GetVoltage(UInt32 alignmentAInCh)
    Double GetPower(UInt32 alignmentPMCh)
    Double GetMeasurementWaveLength(UInt32 alignmentPMCh)
    ErrorCode SetSingle(ISingleParameter parameter)
    ErrorCode SetSingle(ISingleParameter parameter, RotationCenter rotationCenter)
    ErrorCode SetFlat(IFlatParameter parameter)
    ErrorCode SetFlat(IFlatParameter parameter, RotationCenter rotationCenter)
    ErrorCode SetFocus(IFocusParameter parameter)
    ErrorCode SetFocus(IFocusParameter parameter, RotationCenter rotationCenter)
    ErrorCode SetTwoChRotation(ITwoChRotationParameter parameter)
    ErrorCode SetTwoChRotation(ITwoChRotationParameter parameter, RotationCenter rotationCenter)
    ErrorCode SetMMFRotation(IMMFRotationParameter parameter)
    ErrorCode SetMMFRotation(IMMFRotationParameter parameter, RotationCenter rotationCenter)
    ErrorCode SetLoopBack(ILoopBackParameter parameter)
    ErrorCode SetLoopBack(ILoopBackParameter parameter, RotationCenter rotationCenter)
    ErrorCode SetCommon(CommonParameter parameter)
    ErrorCode SetCommon(CommonParameter parameter, RotationCenter rotationCenter)
    ErrorCode SetMeasurementWaveLength(UInt32 alignmentPMCh, Double waveLength)

Tipos anidados/enums:
    SurugaSeiki.Motion.Alignment+AlignmentType
    SurugaSeiki.Motion.Alignment+ErrorCode
    SurugaSeiki.Motion.Alignment+Status
    SurugaSeiki.Motion.Alignment+AligningStatusCode
    SurugaSeiki.Motion.Alignment+ProfileDataType
    SurugaSeiki.Motion.Alignment+ZMode
    SurugaSeiki.Motion.Alignment+ISharingParameter
    SurugaSeiki.Motion.Alignment+ISingleParameter
    SurugaSeiki.Motion.Alignment+SingleParameter
    SurugaSeiki.Motion.Alignment+IFlatParameter
    SurugaSeiki.Motion.Alignment+FlatParameter
    SurugaSeiki.Motion.Alignment+IFocusParameter
    SurugaSeiki.Motion.Alignment+FocusParameter
    SurugaSeiki.Motion.Alignment+ITwoChRotationParameter
    SurugaSeiki.Motion.Alignment+TwoChRotationParameter
    SurugaSeiki.Motion.Alignment+IMMFRotationParameter
    SurugaSeiki.Motion.Alignment+MMFRotationParameter
    SurugaSeiki.Motion.Alignment+ILoopBackParameter
    SurugaSeiki.Motion.Alignment+LoopBackParameter
    SurugaSeiki.Motion.Alignment+ICommonParameter
    SurugaSeiki.Motion.Alignment+CommonParameter
    SurugaSeiki.Motion.Alignment+ProfileData

=====
SurugaSeiki.Motion.Alignment+AligningStatusCode
=====

Campos:
    Int32 value__
    static AligningStatusCode NotAligning
    static AligningStatusCode Initializing
    static AligningStatusCode FieldSearching
    static AligningStatusCode PeakSearchingX
    static AligningStatusCode PeakSearchingY
    static AligningStatusCode PeakSearchingZ
    static AligningStatusCode PeakSearchXCh2
    static AligningStatusCode PeakSearchYCh2

=====
SurugaSeiki.Motion.Alignment+AlignmentType
=====

Campos:
    Int32 value__
    static AlignmentType Single
    static AlignmentType Flat
    static AlignmentType Focus
    static AlignmentType TwoChRotation
    static AlignmentType MMFRotation
    static AlignmentType LoopBack
    static AlignmentType TwoChFlat

=====
SurugaSeiki.Motion.Alignment+CommonParameter
=====

Propiedades:
    AlignmentType alignmentType (get/set)
    ZMode zMode (get/set)
    UInt16 mainStageNumberX (get/set)

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UInt16 mainStageNumberY (get/set)
UInt16 mainStageNumberZ (get/set)
UInt16 subStageNumberX (get/set)
UInt16 subStageNumberXY (get/set)
Double subAngleX (get/set)
Double subAngleY (get/set)
UInt16 rotateStageNumber (get/set)
Double rotateSpeed (get/set)
Int16 rotateDirection (get/set)
Double coreDistance (get/set)
UInt16 mmfStageNumber (get/set)
Double mmfSpeed (get/set)
Double mmfDistance (get/set)
UInt16 pmCh (get/set)
UInt16 pmCh1 (get/set)
UInt16 pmCh2 (get/set)
UInt16 analogCh (get/set)
UInt16 analogCh1 (get/set)
UInt16 analogCh2 (get/set)
Boolean pmAutoRangeUpOn (get/set)
Boolean pmInitRangeSettingOn (get/set)
Int16 pmInitRange (get/set)
Int16 pmInitRangeCh1 (get/set)
Int16 pmInitRangeCh2 (get/set)
Double fieldSearchThreshold (get/set)
Double fieldSearchThresholdCh1 (get/set)
Double fieldSearchThresholdCh2 (get/set)
Double peakSearchThreshold (get/set)
Double peakSearchThresholdXY (get/set)
Double peakSearchThresholdCh1 (get/set)
Double peakSearchThresholdCh2 (get/set)
Double peakSearchThresholdZ (get/set)
Double searchRangeX (get/set)
Double searchRangeY (get/set)
Double searchRangeZ (get/set)
Double fieldSearchPitchX (get/set)
Double fieldSearchPitchY (get/set)
Double fieldSearchPitchZ (get/set)
Double fieldSearchFirstPitchX (get/set)
Double fieldSearchSpeedX (get/set)
Double fieldSearchSpeedY (get/set)
Double fieldSearchSpeedZ (get/set)
Double peakSearchSpeedX (get/set)
Double peakSearchSpeedY (get/set)
Double peakSearchSpeedZ (get/set)
UInt32 smoothingRangeX (get/set)
UInt32 smoothingRangeY (get/set)
UInt32 smoothingRangeZ (get/set)
Double centroidThresholdX (get/set)
Double centroidThresholdY (get/set)
Double centroidThresholdZ (get/set)
Double convergentRangeX (get/set)
Double convergentRangeY (get/set)
Double convergentRangeZ (get/set)
UInt16 comparisonCount (get/set)
UInt16 maxRepeatCount (get/set)

=====
SurugaSeiki.Motion.Alignment+ErrorCode
=====

Campos:
    Int32 value_
    static ErrorCode None
    static ErrorCode Axis
    static ErrorCode Aligning
    static ErrorCode Parameter

=====
SurugaSeiki.Motion.Alignment+FlatParameter
=====

Propiedades:
    UInt16 mainStageNumberX (get/set)
    UInt16 mainStageNumberY (get/set)
    UInt16 subStageNumberXY (get/set)
    Double subAngleX (get/set)
    Double subAngleY (get/set)
    UInt16 pmCh (get/set)
    UInt16 analogCh (get/set)
    Boolean pmAutoRangeUpOn (get/set)
    Boolean pmInitRangeSettingOn (get/set)
    Int16 pmInitRange (get/set)
    Double fieldSearchThreshold (get/set)
    Double peakSearchThreshold (get/set)
    Double searchRangeX (get/set)
    Double searchRangeY (get/set)
    Double fieldSearchPitchX (get/set)
    Double fieldSearchPitchY (get/set)
    Double fieldSearchFirstPitchX (get/set)
    Double fieldSearchSpeedX (get/set)
    Double fieldSearchSpeedY (get/set)
    Double peakSearchSpeedX (get/set)
    Double peakSearchSpeedY (get/set)
    UInt32 smoothingRangeX (get/set)
    UInt32 smoothingRangeY (get/set)
    Double centroidThresholdX (get/set)
    Double centroidThresholdY (get/set)
    Double convergentRangeX (get/set)

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Double convergentRangeY (get/set)
UInt16 comparisonCount (get/set)
UInt16 maxRepeatCount (get/set)

=====
SurugaSeiki.Motion.Alignment+FocusParameter
=====

Propiedades:
ZMode zMode (get/set)
UInt16 mainStageNumberX (get/set)
UInt16 mainStageNumberY (get/set)
UInt16 mainStageNumberZ (get/set)
UInt16 subStageNumberXY (get/set)
Double subAngleX (get/set)
Double subAngleY (get/set)
UInt16 pmCh (get/set)
UInt16 analogCh (get/set)
Boolean pmAutoRangeUpOn (get/set)
Boolean pmInitRangeSettingOn (get/set)
Int16 pmInitRange (get/set)
Double fieldSearchThreshold (get/set)
Double peakSearchThresholdXY (get/set)
Double peakSearchThresholdZ (get/set)
Double searchRangeX (get/set)
Double searchRangeY (get/set)
Double searchRangeZ (get/set)
Double fieldSearchPitchX (get/set)
Double fieldSearchPitchY (get/set)
Double fieldSearchPitchZ (get/set)
Double fieldSearchFirstPitchX (get/set)
Double fieldSearchSpeedX (get/set)
Double fieldSearchSpeedY (get/set)
Double fieldSearchSpeedZ (get/set)
Double peakSearchSpeedX (get/set)
Double peakSearchSpeedY (get/set)
Double peakSearchSpeedZ (get/set)
UInt32 smoothingRangeX (get/set)
UInt32 smoothingRangeY (get/set)
UInt32 smoothingRangeZ (get/set)
Double centroidThresholdX (get/set)
Double centroidThresholdY (get/set)
Double centroidThresholdZ (get/set)
Double convergentRangeX (get/set)
Double convergentRangeY (get/set)
Double convergentRangeZ (get/set)
UInt16 comparisonCount (get/set)
UInt16 maxRepeatCount (get/set)

=====
SurugaSeiki.Motion.Alignment+ICommonParameter
=====

Propiedades:
AlignmentType alignmentType (get/set)

=====
SurugaSeiki.Motion.Alignment+IFlatParameter
=====

Propiedades:
UInt16 mainStageNumberY (get/set)
UInt16 subStageNumberXY (get/set)
Double subAngleY (get/set)
UInt16 pmCh (get/set)
UInt16 analogCh (get/set)
Int16 pmInitRange (get/set)
Double fieldSearchThreshold (get/set)
Double peakSearchThreshold (get/set)
Double searchRangeY (get/set)
Double fieldSearchPitchX (get/set)
Double fieldSearchPitchY (get/set)
Double fieldSearchFirstPitchX (get/set)
Double fieldSearchSpeedY (get/set)
Double peakSearchSpeedY (get/set)
UInt32 smoothingRangeY (get/set)
Double centroidThresholdY (get/set)
Double convergentRangeY (get/set)

=====
SurugaSeiki.Motion.Alignment+IFocusParameter
=====

Propiedades:
ZMode zMode (get/set)
UInt16 mainStageNumberY (get/set)
UInt16 mainStageNumberZ (get/set)
UInt16 subStageNumberXY (get/set)
Double subAngleY (get/set)
UInt16 pmCh (get/set)
UInt16 analogCh (get/set)
Int16 pmInitRange (get/set)
Double fieldSearchThreshold (get/set)
Double peakSearchThresholdXY (get/set)
Double peakSearchThresholdZ (get/set)
Double searchRangeY (get/set)
Double searchRangeZ (get/set)
Double fieldSearchPitchX (get/set)
Double fieldSearchPitchY (get/set)

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Double fieldSearchPitchZ (get/set)
Double fieldSearchFirstPitchX (get/set)
Double fieldSearchSpeedY (get/set)
Double fieldSearchSpeedZ (get/set)
Double peakSearchSpeedY (get/set)
Double peakSearchSpeedZ (get/set)
UInt32 smoothingRangeY (get/set)
UInt32 smoothingRangeZ (get/set)
Double centroidThresholdY (get/set)
Double centroidThresholdZ (get/set)
Double convergentRangeY (get/set)
Double convergentRangeZ (get/set)

=====
SurugaSeiki.Motion.Alignment+ILoopBackParameter
=====

Propiedades:
  UInt16 mainStageNumberY (get/set)
  UInt16 mainStageNumberZ (get/set)
  UInt16 subStageNumberXY (get/set)
  Double subAngleY (get/set)
  UInt16 pmCh (get/set)
  UInt16 analogCh (get/set)
  Int16 pmInitRange (get/set)
  Double fieldSearchThreshold (get/set)
  Double peakSearchThresholdXY (get/set)
  Double peakSearchThresholdZ (get/set)
  Double searchRangeY (get/set)
  Double searchRangeZ (get/set)
  Double fieldSearchPitchX (get/set)
  Double fieldSearchPitchY (get/set)
  Double fieldSearchPitchZ (get/set)
  Double fieldSearchFirstPitchX (get/set)
  Double fieldSearchSpeedY (get/set)
  Double fieldSearchSpeedZ (get/set)
  Double peakSearchSpeedY (get/set)
  Double peakSearchSpeedZ (get/set)
  UInt32 smoothingRangeY (get/set)
  UInt32 smoothingRangeZ (get/set)
  Double centroidThresholdY (get/set)
  Double centroidThresholdZ (get/set)
  Double convergentRangeY (get/set)
  Double convergentRangeZ (get/set)

=====
SurugaSeiki.Motion.Alignment+IMMFRotationParameter
=====

Propiedades:
  UInt16 mmfStageNumber (get/set)
  Double mmfSpeed (get/set)
  Double mmfDistance (get/set)

=====
SurugaSeiki.Motion.Alignment+ISharingParameter
=====

Propiedades:
  UInt16 mainStageNumberX (get/set)
  Double subAngleX (get/set)
  Boolean pmAutoRangeUpOn (get/set)
  Boolean pmInitRangeSettingOn (get/set)
  Double searchRangeX (get/set)
  Double fieldSearchSpeedX (get/set)
  Double peakSearchSpeedX (get/set)
  UInt32 smoothingRangeX (get/set)
  Double centroidThresholdX (get/set)
  Double convergentRangeX (get/set)
  UInt16 comparisonCount (get/set)
  UInt16 maxRepeatCount (get/set)

=====
SurugaSeiki.Motion.Alignment+ISingleParameter
=====

Propiedades:
  ZMode zMode (get/set)
  UInt16 subStageNumberX (get/set)
  UInt16 pmCh (get/set)
  UInt16 analogCh (get/set)
  Int16 pmInitRange (get/set)
  Double fieldSearchThreshold (get/set)
  Double peakSearchThreshold (get/set)

=====
SurugaSeiki.Motion.Alignment+ITwoChRotationParameter
=====

Propiedades:
  UInt16 mainStageNumberY (get/set)
  UInt16 subStageNumberXY (get/set)
  Double subAngleY (get/set)
  UInt16 rotateStageNumber (get/set)
  Double rotateSpeed (get/set)
  Int16 rotateDirection (get/set)
  Double coreDistance (get/set)
  UInt16 pmCh1 (get/set)
  UInt16 pmCh2 (get/set)

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UInt16 analogCh1 (get/set)
UInt16 analogCh2 (get/set)
Int16 pmInitRangeCh1 (get/set)
Int16 pmInitRangeCh2 (get/set)
Double fieldSearchThresholdCh1 (get/set)
Double fieldSearchThresholdCh2 (get/set)
Double peakSearchThresholdCh1 (get/set)
Double peakSearchThresholdCh2 (get/set)
Double searchRangeY (get/set)
Double fieldSearchPitchX (get/set)
Double fieldSearchPitchY (get/set)
Double fieldSearchFirstPitchX (get/set)
Double fieldSearchSpeedY (get/set)
Double peakSearchSpeedY (get/set)
UInt32 smoothingRangeY (get/set)
Double centroidThresholdY (get/set)
Double convergentRangeY (get/set)

=====
SurugaSeiki.Motion.Alignment+LoopBackParameter
=====

Propiedades:
  UInt16 mainStageNumberX (get/set)
  UInt16 mainStageNumberY (get/set)
  UInt16 mainStageNumberZ (get/set)
  UInt16 subStageNumberXY (get/set)
  Double subAngleX (get/set)
  Double subAngleY (get/set)
  UInt16 pmCh (get/set)
  UInt16 analogCh (get/set)
  Boolean pmAutoRangeUpOn (get/set)
  Boolean pmInitRangeSettingOn (get/set)
  Int16 pmInitRange (get/set)
  Double fieldSearchThreshold (get/set)
  Double peakSearchThresholdXY (get/set)
  Double peakSearchThresholdZ (get/set)
  Double searchRangeX (get/set)
  Double searchRangeY (get/set)
  Double searchRangeZ (get/set)
  Double fieldSearchPitchX (get/set)
  Double fieldSearchPitchY (get/set)
  Double fieldSearchPitchZ (get/set)
  Double fieldSearchFirstPitchX (get/set)
  Double fieldSearchSpeedX (get/set)
  Double fieldSearchSpeedY (get/set)
  Double fieldSearchSpeedZ (get/set)
  Double peakSearchSpeedX (get/set)
  Double peakSearchSpeedY (get/set)
  Double peakSearchSpeedZ (get/set)
  UInt32 smoothingRangeX (get/set)
  UInt32 smoothingRangeY (get/set)
  UInt32 smoothingRangeZ (get/set)
  Double centroidThresholdX (get/set)
  Double centroidThresholdY (get/set)
  Double centroidThresholdZ (get/set)
  Double convergentRangeX (get/set)
  Double convergentRangeY (get/set)
  Double convergentRangeZ (get/set)
  UInt16 comparisonCount (get/set)
  UInt16 maxRepeatCount (get/set)

=====
SurugaSeiki.Motion.Alignment+MMFRotationParameter
=====

Propiedades:
  UInt16 mainStageNumberX (get/set)
  UInt16 mainStageNumberY (get/set)
  UInt16 subStageNumberXY (get/set)
  Double subAngleX (get/set)
  Double subAngleY (get/set)
  UInt16 rotateStageNumber (get/set)
  Double rotateSpeed (get/set)
  Int16 rotateDirection (get/set)
  Double coreDistance (get/set)
  UInt16 mmfStageNumber (get/set)
  Double mmfSpeed (get/set)
  Double mmfDistance (get/set)
  UInt16 pmCh1 (get/set)
  UInt16 pmCh2 (get/set)
  UInt16 analogCh1 (get/set)
  UInt16 analogCh2 (get/set)
  Boolean pmAutoRangeUpOn (get/set)
  Boolean pmInitRangeSettingOn (get/set)
  Int16 pmInitRangeCh1 (get/set)
  Int16 pmInitRangeCh2 (get/set)
  Double fieldSearchThresholdCh1 (get/set)
  Double fieldSearchThresholdCh2 (get/set)
  Double peakSearchThresholdCh1 (get/set)
  Double peakSearchThresholdCh2 (get/set)
  Double searchRangeX (get/set)
  Double searchRangeY (get/set)
  Double fieldSearchPitchX (get/set)
  Double fieldSearchPitchY (get/set)
  Double fieldSearchFirstPitchX (get/set)
  Double fieldSearchSpeedX (get/set)
  Double fieldSearchSpeedY (get/set)
  Double peakSearchSpeedX (get/set)

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Double peakSearchSpeedY (get/set)
UInt32 smoothingRangeX (get/set)
UInt32 smoothingRangeY (get/set)
Double centroidThresholdX (get/set)
Double centroidThresholdY (get/set)
Double convergentRangeX (get/set)
Double convergentRangeY (get/set)
UInt16 comparisonCount (get/set)
UInt16 maxRepeatCount (get/set)

=====
SurugaSeiki.Motion.Alignment+ProfileData
=====

Campos:
  UInt16 mainAxisNumber
  UInt16 sub1AxisNumber
  UInt16 sub2AxisNumber
  UInt16 signalCh1Number
  UInt16 signalCh2Number
  UInt32 packetIndex
  UInt32 dataCount
  Double[] mainPositionList
  Double[] sub1PositionList
  Double[] sub2PositionList
  Double[] signalCh1List
  Double[] signalCh2List

=====
SurugaSeiki.Motion.Alignment+ProfileDataType
=====

Campos:
  Int32 value_
  static ProfileDataType FieldSearch
  static ProfileDataType PeakSearchX
  static ProfileDataType PeakSearchY
  static ProfileDataType PeakSearchZ
  static ProfileDataType PeakSearchXCh2
  static ProfileDataType PeakSearchYCh2

=====
SurugaSeiki.Motion.Alignment+SingleParameter
=====

Propiedades:
  ZMode zMode (get/set)
  UInt16 mainStageNumberX (get/set)
  UInt16 subStageNumberX (get/set)
  Double subAngleX (get/set)
  UInt16 pmCh (get/set)
  UInt16 analogCh (get/set)
  Boolean pmAutoRangeUpOn (get/set)
  Boolean pmInitRangeSettingOn (get/set)
  Int16 pmInitRange (get/set)
  Double fieldSearchThreshold (get/set)
  Double peakSearchThreshold (get/set)
  Double searchRangeX (get/set)
  Double fieldSearchSpeedX (get/set)
  Double peakSearchSpeedX (get/set)
  UInt32 smoothingRangeX (get/set)
  Double centroidThresholdX (get/set)
  Double convergentRangeX (get/set)
  UInt16 comparisonCount (get/set)
  UInt16 maxRepeatCount (get/set)

=====
SurugaSeiki.Motion.Alignment+Status
=====

Campos:
  Int32 value_
  static Status Stopping
  static Status Success
  static Status Aligning
  static Status FieldSearchRangeOver
  static Status ProfileDataOver
  static Status PeakSearchCountOver
  static Status PeakSearchRangeOver
  static Status InvalidParameter
  static Status ServoIsNotReady
  static Status ServoIsAlarm
  static Status StageOnLimit
  static Status VoltageLimit
  static Status PMRangeLimit
  static Status PMInitRangeChangeFail
  static Status PMDisconnected
  static Status RotationAdjustmentFail
  static Status InPositionFail
  static Status TorqueLimit

=====
SurugaSeiki.Motion.Alignment+TwoChRotationParameter
=====

Propiedades:
  UInt16 mainStageNumberX (get/set)
  UInt16 mainStageNumberY (get/set)
  UInt16 subStageNumberXY (get/set)

```

```

Double subAngleX (get/set)
Double subAngleY (get/set)
UInt16 rotateStageNumber (get/set)
Double rotateSpeed (get/set)
Int16 rotateDirection (get/set)
Double coreDistance (get/set)
UInt16 pmCh1 (get/set)
UInt16 pmCh2 (get/set)
UInt16 analogCh1 (get/set)
UInt16 analogCh2 (get/set)
Boolean pmAutoRangeUpOn (get/set)
Boolean pmInitRangeSettingOn (get/set)
Int16 pmInitRangeCh1 (get/set)
Int16 pmInitRangeCh2 (get/set)
Double fieldSearchThresholdCh1 (get/set)
Double fieldSearchThresholdCh2 (get/set)
Double peakSearchThresholdCh1 (get/set)
Double peakSearchThresholdCh2 (get/set)
Double searchRangeX (get/set)
Double searchRangeY (get/set)
Double fieldSearchPitchX (get/set)
Double fieldSearchPitchY (get/set)
Double fieldSearchFirstPitchX (get/set)
Double fieldSearchSpeedX (get/set)
Double fieldSearchSpeedY (get/set)
Double peakSearchSpeedX (get/set)
Double peakSearchSpeedY (get/set)
UInt32 smoothingRangeX (get/set)
UInt32 smoothingRangeY (get/set)
Double centroidThresholdX (get/set)
Double centroidThresholdY (get/set)
Double convergentRangeX (get/set)
Double convergentRangeY (get/set)
UInt16 comparisonCount (get/set)
UInt16 maxRepeatCount (get/set)

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```

=====
SurugaSeiki.Motion.Alignment+ZMode
=====

```

```

Campos:
    Int32 value_
    static ZMode Round
    static ZMode ForwardPlus
    static ZMode ForwardMinus

```

3.2 Clase AngleAdjustment

```

=====
SurugaSeiki.Motion.AngleAdjustment
=====

```

```

Constructores:
    AngleAdjustment(UInt16 id)

Metodos:
    ErrorCode Start()
    ErrorCode Stop()
    Status GetStatus()
    UInt32 GetErrorAxisID()
    AdjustingStatus GetAdjustingStatus()
    UInt32 GetProfilePacketSumIndex(ProfileDataType profileDataType)
    ProfileData RequestProfileData(ProfileDataType profileDataType, UInt32 index)
    ErrorCode SetParameter(AngleAdjustmentParameter parameter)
    ErrorCode SetParameter(AngleAdjustmentParameter parameter, RotationCenter rotationCenter)

```

```

Tipos anidados/enums:
    SurugaSeiki.Motion.AngleAdjustment+ErrorCode
    SurugaSeiki.Motion.AngleAdjustment+Status
    SurugaSeiki.Motion.AngleAdjustment+AdjustingStatus
    SurugaSeiki.Motion.AngleAdjustment+ProfileDataType
    SurugaSeiki.Motion.AngleAdjustment+AngleAdjustmentParameter
    SurugaSeiki.Motion.AngleAdjustment+ProfileData

```

```

=====
SurugaSeiki.Motion.AngleAdjustment+AdjustingStatus
=====

```

```

Campos:
    Int32 value_
    static AdjustingStatus NotAdjusting
    static AdjustingStatus Initializing
    static AdjustingStatus ContactingZ
    static AdjustingStatus AdjustingTx
    static AdjustingStatus AdjustingTy

```

```

=====
SurugaSeiki.Motion.AngleAdjustment+AngleAdjustmentParameter
=====

```

```

Campos:
    Double gap
    UInt16 signalChNumber

```

```

Double signalLowerLimit
UInt16 unlockDOutChNumber
Boolean unlockDOutControlOn
Boolean lockUnlockAdjustEnbale
Double lockUnlockDifference
UInt16 contactAxisNumber
Double contactSearchRange
Double contactSearchSpeed
UInt32 contactSmoothing
UInt32 contactSensitivity
Double pushDistance
UInt16 angleAxisNumberTx
UInt16 angleAxisNumberTy
Double angleSearchRangeTx
Double angleSearchRangeTy
Double angleSearchSpeedTx
Double angleSearchSpeedTy
UInt32 angleSmoothingTx
UInt32 angleSmoothingTy
UInt32 angleSensitivityTx
UInt32 angleSensitivityTy
UInt16 angleJudgeCountTx
UInt16 angleJudgeCountTy
Double angleConvergentRangeTx
Double angleConvergentRangeTy
UInt16 angleComparisonCount
UInt16 angleMaxCount

=====
SurugaSeiki.Motion.AngleAdjustment+ErrorCode
=====

Campos:
    Int32 value__
    static ErrorCode None
    static ErrorCode Axis
    static ErrorCode Adjusting
    static ErrorCode Parameter

=====
SurugaSeiki.Motion.AngleAdjustment+ProfileData
=====

Campos:
    UInt16 mainAxisNumber
    UInt16 sub1AxisNumber
    UInt16 sub2AxisNumber
    UInt16 signalCh1Number
    UInt16 signalCh2Number
    UInt32 packetIndex
    UInt32 dataCount
    Double[] mainPositionList
    Double[] sub1PositionList
    Double[] sub2PositionList
    Double[] signalCh1List
    Double[] signalCh2List

=====
SurugaSeiki.Motion.AngleAdjustment+ProfileDataType
=====

Campos:
    Int32 value__
    static ProfileDataType ContactZ
    static ProfileDataType AdjustmentTx
    static ProfileDataType AdjustmentTy

=====
SurugaSeiki.Motion.AngleAdjustment+Status
=====

Campos:
    Int32 value__
    static Status Stopping
    static Status Success
    static Status Adjusting
    static Status ProfileDataOver
    static Status InvalidParameter
    static Status ServoIsNotReady
    static Status ServoIsAlarm
    static Status StageOnLimit
    static Status SignalLowerLimit
    static Status CouldnotContact
    static Status AdjustCountOver
    static Status AngleAdjustRnageOver
    static Status LostContact

```

3.3 Clase Axis2D

```

=====
SurugaSeiki.Motion.Axis2D
=====

```

```
Constructores:
    Axis2D(UInt16 mainStageNumber, UInt16 subStageNumber)

Metodos:
    ErrorCode SetAxisNumber(UInt16 mainStageNumber, UInt16 subStageNumber)
    UInt16[] GetAxisNumber()
    ErrorCode MoveAbsolute(Point targetPosition)
    ErrorCode MoveAbsolute(Double mainTargetPosition, Double subTargetPosition)
    ErrorCode MoveRelative(Point targetDistance)
    ErrorCode MoveRelative(Double mainDistance, Double subDistance)
    ErrorCode MoveAbsoluteAngle(Double mainTargetPosition, Double subAngle)
    ErrorCode MoveRelativeAngle(Double mainDistance, Double subAngle)
    ErrorCode Stop()
    Point GetTargetPosition()
    Point GetActualPosition()
    Boolean IsMoving()
    ErrorCode SetSpeed(Double speed)
    ErrorCode SetAccelRate(Double rate)
    ErrorCode SetDecelRate(Double rate)
    ErrorCode GetErrorCode()
    Status GetStatus()
```

```
Tipos anidados/enums:
    SurugaSeiki.Motion.Axis2D+Point
```

```
=====
SurugaSeiki.Motion.Axis2D+Point
=====
```

```
Campos:
    Double X
    Double Y
```

3.4 Clase Axis3D

```
=====
SurugaSeiki.Motion.Axis3D
=====
```

```
Constructores:
    Axis3D(UInt16 mainStageNumberX, UInt16 mainStageNumberY, UInt16 mainStageNumberZ)
```

```
Metodos:
    ErrorCode SetAxisNumber(UInt16 mainStageNumberX, UInt16 mainStageNumberY, UInt16 mainStageNumberZ)
    UInt16[] GetAxisNumber()
    ErrorCode MoveAbsolute(Point3D targetPosition)
    ErrorCode MoveAbsolute(Double targetPositionX, Double targetPositionY, Double targetPositionZ)
    ErrorCode MoveAbsolute(Point3D targetPosition, UInt16 dOutId, Int16 dOutOnDelayTime, Int16 dOutOffDelayTime)
    ErrorCode MoveAbsolute(Double targetPositionX, Double targetPositionY, Double targetPositionZ, UInt16 dOutId, Int16 dOutOnDelayTime, Int16 dOutOffDelayTime)
    ErrorCode MoveRelative(Point3D targetDistance)
    ErrorCode MoveRelative(Double targetDistanceX, Double targetDistanceY, Double targetDistanceZ)
    ErrorCode MoveRelative(Point3D targetDistance, UInt16 dOutId, Int16 dOutOnDelayTime, Int16 dOutOffDelayTime)
    ErrorCode MoveRelative(Double targetDistanceX, Double targetDistanceY, Double targetDistanceZ, UInt16 dOutId, Int16 dOutOnDelayTime, Int16 dOutOffDelayTime)
    ErrorCode Stop()
    Point3D GetTargetPosition()
    Point3D GetActualPosition()
    Boolean IsMoving()
    ErrorCode SetSpeed(Double speed)
    ErrorCode SetAccelRate(Double accelRate)
    ErrorCode SetDecelRate(Double decelRate)
    ErrorCode SetStageNumber(UInt16 mainStageNumberX, UInt16 mainStageNumberY, UInt16 mainStageNumberZ)
    ErrorCode SetSubStageNumber(UInt16 subStageNumberX, UInt16 subStageNumberY, UInt16 subStageNumberZ)
    ErrorCode SetRotationCenter(Double rotationOffsetX, Double rotationOffsetY, Double rotationOffsetZ)
    ErrorCode GetErrorCode()
    Status GetStatus()
```

```
Tipos anidados/enums:
    SurugaSeiki.Motion.Axis3D+Point3D
    SurugaSeiki.Motion.Axis3D+RotationCenter
```

```
=====
SurugaSeiki.Motion.Axis3D+Point3D
=====
```

```
Campos:
    Double X
    Double Y
    Double Z
```

```
=====
SurugaSeiki.Motion.Axis3D+RotationCenter
=====
```

```
Campos:
    UInt16 mainStageNumberX
    UInt16 mainStageNumberY
    UInt16 mainStageNumberZ
    UInt16 subStageNumberX
    UInt16 subStageNumberY
    UInt16 subStageNumberZ
    Double offsetX
```

```
Double offsetY
Double offsetZ
```

3.5 Clase AxisComponents

```
=====
SurugaSeiki.Motion.AxisComponents
=====

Constructores:
    AxisComponents(UInt16 axisNumbern)

Metodos:
    ErrorCode MoveAbsolute(Double position)
    ErrorCode MoveRelative(Double distance)
    ErrorCode ReturnOrigin()
    ErrorCode Stop()
    ErrorCode TurnOnServo()
    ErrorCode TurnOffServo()
    ErrorCode ResetError()
    Double GetTargetPosition()
    Double GetActualPosition()
    Boolean IsServoOn()
    Boolean IsServoAlarm()
    Boolean IsMoving()
    Boolean IsOriginSignalOn()
    Boolean IsCwLimitSignalOn()
    Boolean IsCcwLimitSignalOn()
    Boolean IsCwTorqueLimitOn()
    Boolean IsCcwTorqueLimitOn()
    Boolean IsOriginReturnSucceeded()
    ErrorCode SetAccelRate(Double rate)
    ErrorCode SetDecelRate(Double rate)
    ErrorCode SetSineMotion(Boolean sineMotion)
    ErrorCode SetFulcrumSpan(Double span)
    ErrorCode SetEncoderStepLength(Double length)
    ErrorCode SetMaxSpeed(Double speed)
    ErrorCode SetCWTorqueLimit(Double cWlimit)
    ErrorCode SetCCWTorqueLimit(Double cCWlimit)
    ErrorCode GetErrorCode()
    Status GetStatus()

Tipos anidados/enums:
    SurugaSeiki.Motion.AxisComponents+ErrorCode
    SurugaSeiki.Motion.AxisComponents+Status
```

```
=====
SurugaSeiki.Motion.AxisComponents+ErrorCode
=====
```

```
Campos:
    Int32 value__
    static ErrorCode None
    static ErrorCode ServoAlarm
    static ErrorCode ServoOff
    static ErrorCode Moving
    static ErrorCode NotConnected
    static ErrorCode Limit
    static ErrorCode Origin
    static ErrorCode Parameter
    static ErrorCode Torque
```

```
=====
SurugaSeiki.Motion.AxisComponents+Status
=====
```

```
Campos:
    Int32 value__
    static Status ServoAlarm
    static Status ServoOff
    static Status Stopping
    static Status InPosition
    static Status Moving
    static Status NotConnected
    static Status CwLimit
    static Status CcwLimit
    static Status CwTorqueLimit
    static Status CcwTorqueLimit
```

3.6 Clase ErrorLog

```
=====
SurugaSeiki.Motion.ErrorLog
=====
```

3.7 Clase IO

```

=====
SurugaSeiki.Motion.IO
=====

Constructores:
    IO()

Metodos:
    Boolean GetPortState(DigitalIOType iOType, UInt16 id)
    Void SetPortState(UInt16 id, Boolean portState)
    Void SetPulseOutput(UInt16 id, UInt16 delayTime, UInt16 highWidth, UInt16 lowWidth, Int16 repetitionCount)
    Double GetAnalogValue(AnalogIOType iOType, UInt16 id)
    Void SetAnalogValue(UInt16 id, Double setValue)

Campos:
    static Int16 RepetitionCountForever

Tipos anidados/enums:
    SurugaSeiki.Motion.IO+DigitalIOType
    SurugaSeiki.Motion.IO+AnalogIOType

=====
SurugaSeiki.Motion.IO+AnalogIOType
=====

Campos:
    Int32 value_
    static AnalogIOType Input
    static AnalogIOType Output

=====
SurugaSeiki.Motion.IO+DOutControlParameterToAlignmentSystem
=====

=====
SurugaSeiki.Motion.IO+DigitalIOType
=====

Campos:
    Int32 value_
    static DigitalIOType Input
    static DigitalIOType Output

```

3.8 Clase PowerMeter

```

=====
SurugaSeiki.Motion.PowerMeter
=====

Constructores:
    PowerMeter()

Metodos:
    Double GetPower(UInt16 ch)
    AllowedRange GetRange(UInt16 ch)
    Double GetWavelength(UInt16 ch)
    ErrorCode SetRange(UInt16 ch, AllowedRange range)
    ErrorCode SetWaveLength(UInt16 ch, Double waveLength)
    ErrorCode PauseCommunication(UInt16 ch, Boolean isEnabled)

Tipos anidados/enums:
    SurugaSeiki.Motion.PowerMeter+ErrorCode
    SurugaSeiki.Motion.PowerMeter+AllowedRange

=====
SurugaSeiki.Motion.PowerMeter+AllowedRange
=====

Campos:
    Int16 value_
    static AllowedRange Minus60
    static AllowedRange Minus50
    static AllowedRange Minus40
    static AllowedRange Minus30
    static AllowedRange Minus20
    static AllowedRange Minus10
    static AllowedRange Zero
    static AllowedRange Plus10
    static AllowedRange Plus20
    static AllowedRange Auto
    static AllowedRange Invalid

=====
SurugaSeiki.Motion.PowerMeter+ErrorCode
=====

Campos:
    Int32 value_
    static ErrorCode None
    static ErrorCode ChannelOutOfRange

```

```
static ErrorCode Invalid
```

3.9 Clase System

```
=====
SurugaSeiki.Motion.System
=====
```

Propiedades:

```
UInt16 MaxAxisStageNumber (get/set)
UInt16 MaxAnalogInputChNumber (get/set)
UInt16 MaxAlignmentNumber (get/set)
UInt16 MaxAngleAdjustmentNumber (get/set)
UInt16 MaxDInChNumber (get/set)
UInt16 MaxDOutChNumber (get/set)
UInt16 MaxAnalogOutputChNumber (get/set)
UInt16 MaxPowerMeterChNumber (get/set)
System Instance (get)
Boolean Connected (get)
String DllVersion (get)
String SystemVersion (get)
```

Metodos:

```
Boolean IsError()
ErrorCode GetCurrentErrorCode()
Boolean IsEmergencyAsserted()
Void SetAddress(String addr)
Void DisconnectAds()
```

Campos:

```
static UInt16 MaxProfileNumber
static UInt32 ProfileDataPacketMaxDataNumber
```

Tipos anidados/enums:

```
SurugaSeiki.Motion.System+ErrorCode
```

```
=====
SurugaSeiki.Motion.System+AlignmentChParam
=====
```

```
=====
SurugaSeiki.Motion.System+AlignmentCommand
=====
```

Campos:

```
Int32 value__
static AlignmentCommand Null
static AlignmentCommand StartSingle
static AlignmentCommand StartFlat
static AlignmentCommand StartFocus
static AlignmentCommand StartTwoChRotation
static AlignmentCommand StartMMFRotation
static AlignmentCommand Stop
static AlignmentCommand SetSingle
static AlignmentCommand SetFlat
static AlignmentCommand SetFocus
static AlignmentCommand SetTwoChRotation
static AlignmentCommand SetMMFRotation
static AlignmentCommand StartLoopBack
static AlignmentCommand SetLoopBack
```

```
=====
SurugaSeiki.Motion.System+AlignmentMainAxisParam
=====
```

```
=====
SurugaSeiki.Motion.System+AlignmentParam
=====
```

```
=====
SurugaSeiki.Motion.System+AlignmentPowerMeterInfo
=====
```

```
=====
SurugaSeiki.Motion.System+AlignmentStatusInfo
=====
```

```
=====
SurugaSeiki.Motion.System+AnalogOutputControlCommand
=====
```

Campos:

```
Int32 value__
static AnalogOutputControlCommand Null
static AnalogOutputControlCommand SetAnalogOutput
```

```
=====
SurugaSeiki.Motion.System+AngleAdjustmentCommand
=====
```

Campos:

```
Int32 value__
static AngleAdjustmentCommand Null
```



```

static AngleAdjustmentCommand SetParameter
static AngleAdjustmentCommand Start
static AngleAdjustmentCommand Stop

=====
SurugaSeiki.Motion.System+AngleAdjustmentParam
=====

SurugaSeiki.Motion.System+AngleAdjustmentStatusInfo
=====

SurugaSeiki.Motion.System+Axis2DCommand
=====

Campos:
    Int32 value__
    static Axis2DCommand Null
    static Axis2DCommand MoveAbsolute
    static Axis2DCommand MoveRelative
    static Axis2DCommand MoveAbsoluteAngle
    static Axis2DCommand MoveRelativeAngle
    static Axis2DCommand Stop
    static Axis2DCommand SetCompositionSpeed
    static Axis2DCommand SetAccelRate
    static Axis2DCommand SetDecelRate
    static Axis2DCommand SetStageNumber

=====
SurugaSeiki.Motion.System+Axis2DParam
=====

SurugaSeiki.Motion.System+AxisCommandStruct
=====

SurugaSeiki.Motion.System+AxisComponentCommand
=====

Campos:
    Int32 value__
    static AxisComponentCommand Null
    static AxisComponentCommand MoveAbsolute
    static AxisComponentCommand MoveRelative
    static AxisComponentCommand ReturnOrigin
    static AxisComponentCommand Stop
    static AxisComponentCommand TurnOnServo
    static AxisComponentCommand TurnOffServo
    static AxisComponentCommand ResetError
    static AxisComponentCommand SetMaxSpeed
    static AxisComponentCommand SetAccelRate
    static AxisComponentCommand SetDecelRate
    static AxisComponentCommand SetSineMotion
    static AxisComponentCommand SetFulcrumSpan
    static AxisComponentCommand SetEncoderStepLength
    static AxisComponentCommand SetCWTorquelimit
    static AxisComponentCommand SetCCWTorquelimit

=====
SurugaSeiki.Motion.System+AxisControlType
=====

Campos:
    Int32 value__
    static AxisControlType AxisComponents
    static AxisControlType Axis2D
    static AxisControlType Axis3D

=====
SurugaSeiki.Motion.System+AxisMethod
=====

Campos:
    Int32 value__
    static AxisMethod Null
    static AxisMethod MoveAbsolute
    static AxisMethod MoveRelative
    static AxisMethod MoveAbsoluteAngle
    static AxisMethod MoveRelativeAngle
    static AxisMethod Stop
    static AxisMethod SetCompositionSpeed
    static AxisMethod SetAccelRate
    static AxisMethod SetDecelRate
    static AxisMethod SetStageNumber
    static AxisMethod SetSubStageNumber
    static AxisMethod SetRotationCenter
    static AxisMethod SetFixedAxisOffset

=====
SurugaSeiki.Motion.System+AxisStatus
=====

SurugaSeiki.Motion.System+DOutControlCommand
=====

Campos:

```

```

    Int32 value__
    static DOutControlCommand Null
    static DOutControlCommand SetDout
    static DOutControlCommand SetPulseOutput

=====
SurugaSeiki.Motion.System+DeviceControlCommand
=====

Campos:
    Int32 value__
    static DeviceControlCommand Null
    static DeviceControlCommand SetWaveLength

=====
SurugaSeiki.Motion.System+DeviceStatusMap
=====

=====
SurugaSeiki.Motion.System+ErrorCode
=====

Campos:
    Int32 value__
    static ErrorCode NoError
    static ErrorCode ConnectionLost
    static ErrorCode VersionError
    static ErrorCode MaxNumberOfErrorCode

=====
SurugaSeiki.Motion.System+Point2d
=====

=====
SurugaSeiki.Motion.System+PowerMeterControlCommand
=====

Campos:
    Int32 value__
    static PowerMeterControlCommand None
    static PowerMeterControlCommand SetRange
    static PowerMeterControlCommand SetWaveLength
    static PowerMeterControlCommand SetCommunicationPaused

=====
SurugaSeiki.Motion.System+ProfileCommand
=====

Campos:
    Int32 value__
    static ProfileCommand Null
    static ProfileCommand SetParameter
    static ProfileCommand Start
    static ProfileCommand Stop

=====
SurugaSeiki.Motion.System+ProfileParam
=====

=====
SurugaSeiki.Motion.System+RotationCenterParam
=====

=====
SurugaSeiki.Motion.System+SystemInfo
=====

```